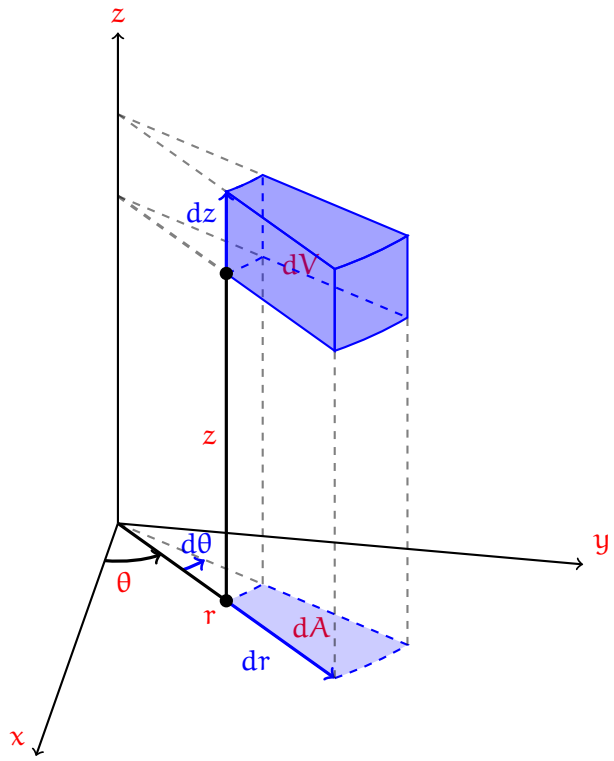
A. **Cylindrical Integration.** We discuss integration in cylindrical coordinates.

Obtain infinitesimal volume dV from the point with coordinates r, θ, z by

- increasing z by:
- increasing θ by:
- increasing r by:

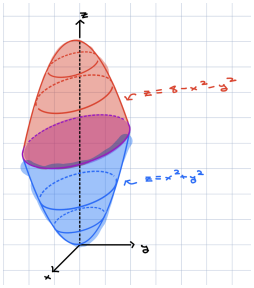


This can be approximated by a rectangular box with

- [base] =
- [height] =

In cylindrical coordinates $dV =$

Example 1. Let E be bounded by paraboloids $z = 8 - x^2 - y^2$ and $z = x^2 + y^2$.



Find $\iiint_E z \, dV$.

Solution. First we describe the paraboloids, and their intersection, in cylindrical.

$z = x^2 + y^2$

$z = 8 - x^2 - y^2$

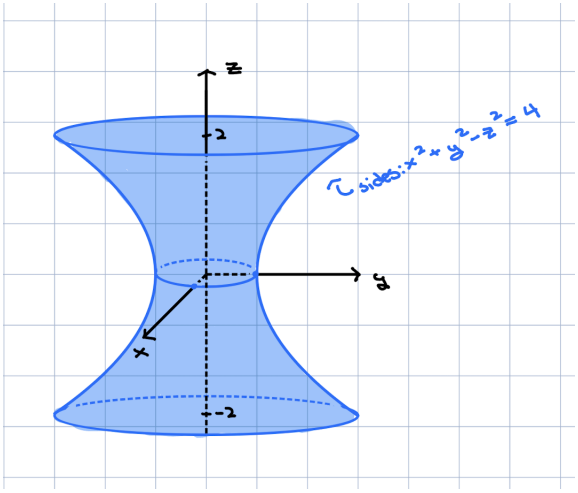
intersection:

The $r\theta$–shadow is described by:

For each fixed (r, θ) from the $r\theta$–shadow, the range of z is:

Thus: $\iiint_E z \, dV =$

Example 2. Use an integral in order $drdz d\theta$ to find the volume of region E bounded by the one–sheeted hyperboloid $x^2 + y^2 - z^2 = 4$ with $-2 \leqslant z \leqslant 2$.



Solution. First we convert the hyperboloid equation to cylindrical.

$x^2 + y^2 - z^2 = 4$

The outer integration variables are $dz d\theta$.

Thus we consider the θz–shadow.

For each fixed (θ, z) from the θz–shadow, the range of r is:

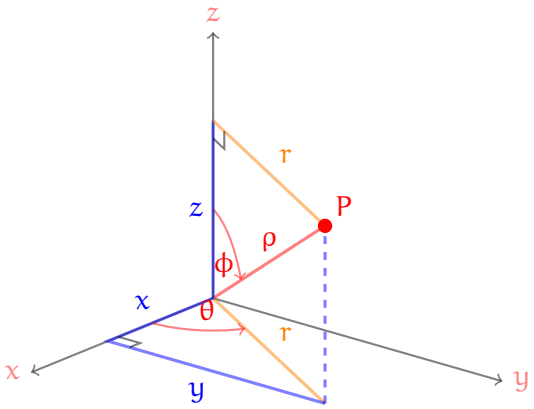
Thus: [volume of E] $=$

B. **Spherical Coordinates.** The **spherical coordinates** of (x, y, z) are ρ, ϕ, and θ.

- when nonnegative, ρ ("rho") measures:
- θ is:
- ϕ ("phi") measures:

and the natural range of ϕ is:

Let's explore in Desmos.



For example we sketch the surface S defined by $\rho = 1$, $0 \leqslant \phi \leqslant \frac{\pi}{2}$, $0 \leqslant \theta \leqslant \frac{\pi}{2}$.

